

ENGINEERING EXPLORATION

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CSE

ENGINEERING

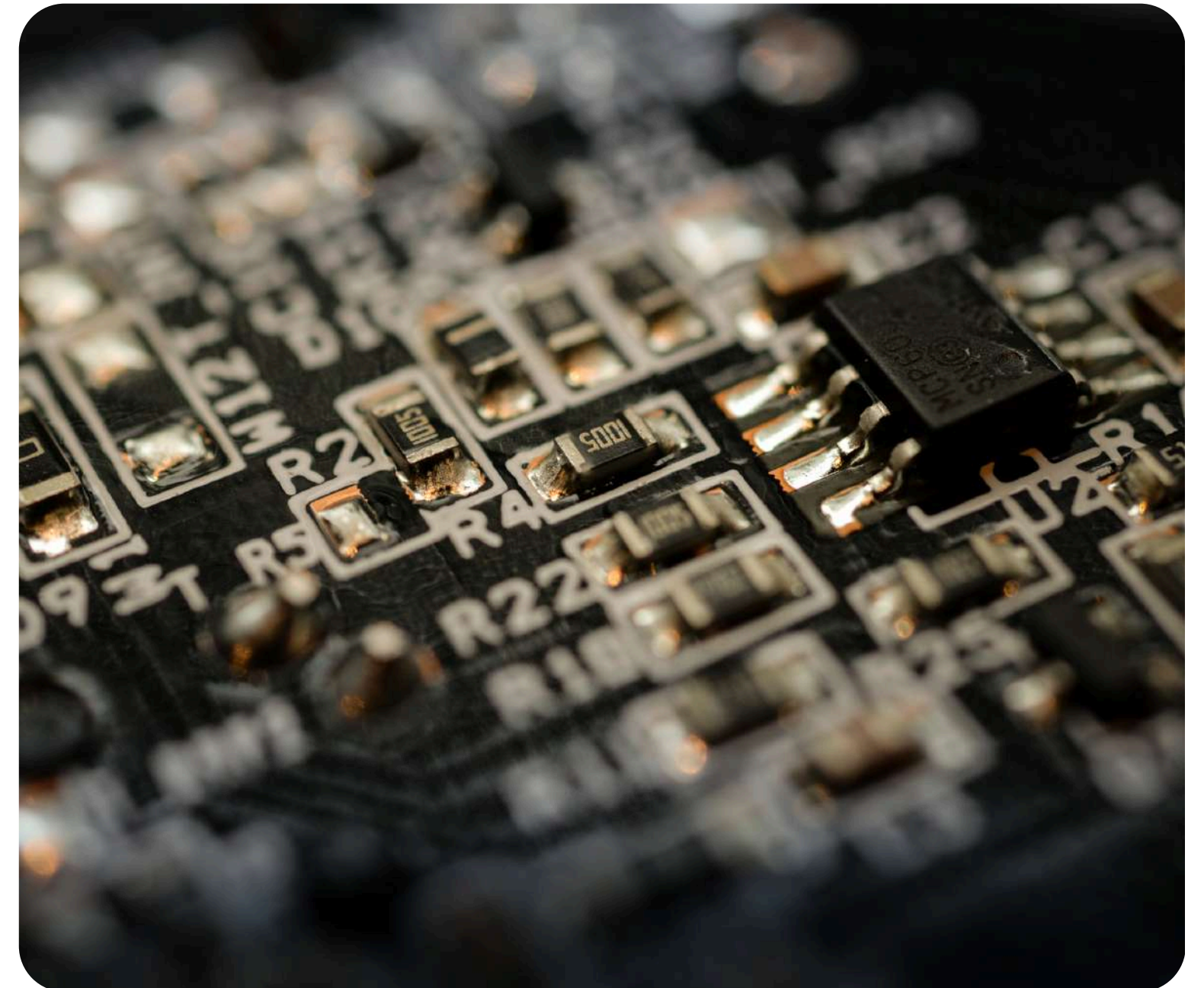
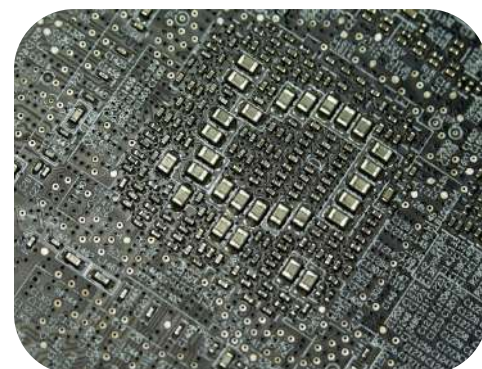
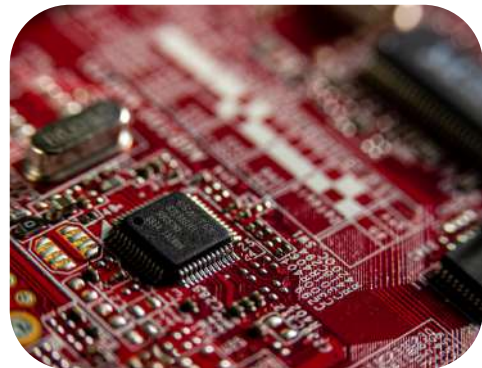


Engineering is the application of scientific and mathematical principles to design, build, improve, and maintain machines, structures, systems, and processes that solve real-world problems.

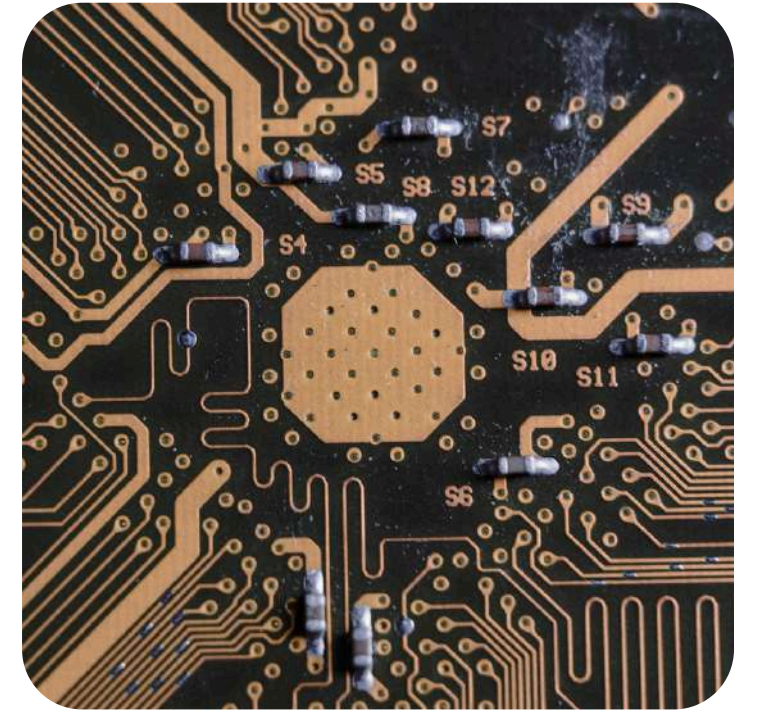
Engineers use creativity and technical knowledge to create solutions that are safe, efficient, and practical.

For example, engineers design:

- * Bridges and buildings
- * Cars and airplanes
- * Computers and smartphones
- * Medical devices



SCIENCE AND ENGINEERING



Science tries to understand how the world works.

Focuses on discovering facts, laws, and theories.

Asks "Why does this happen?"

Uses observation, experiments, and research.



Engineering uses that knowledge to solve real-world problems.

Focuses on designing, building, and improving products or systems.

Asks "How can we use this?"

Uses science, mathematics, and creativity to create practical solutions.

SCIENTIST AND ENGINEER



Studies and discovers
how nature works.

Conducts experiments and
research.

Asks "Why does this happen?"

Creates new knowledge and
theories.

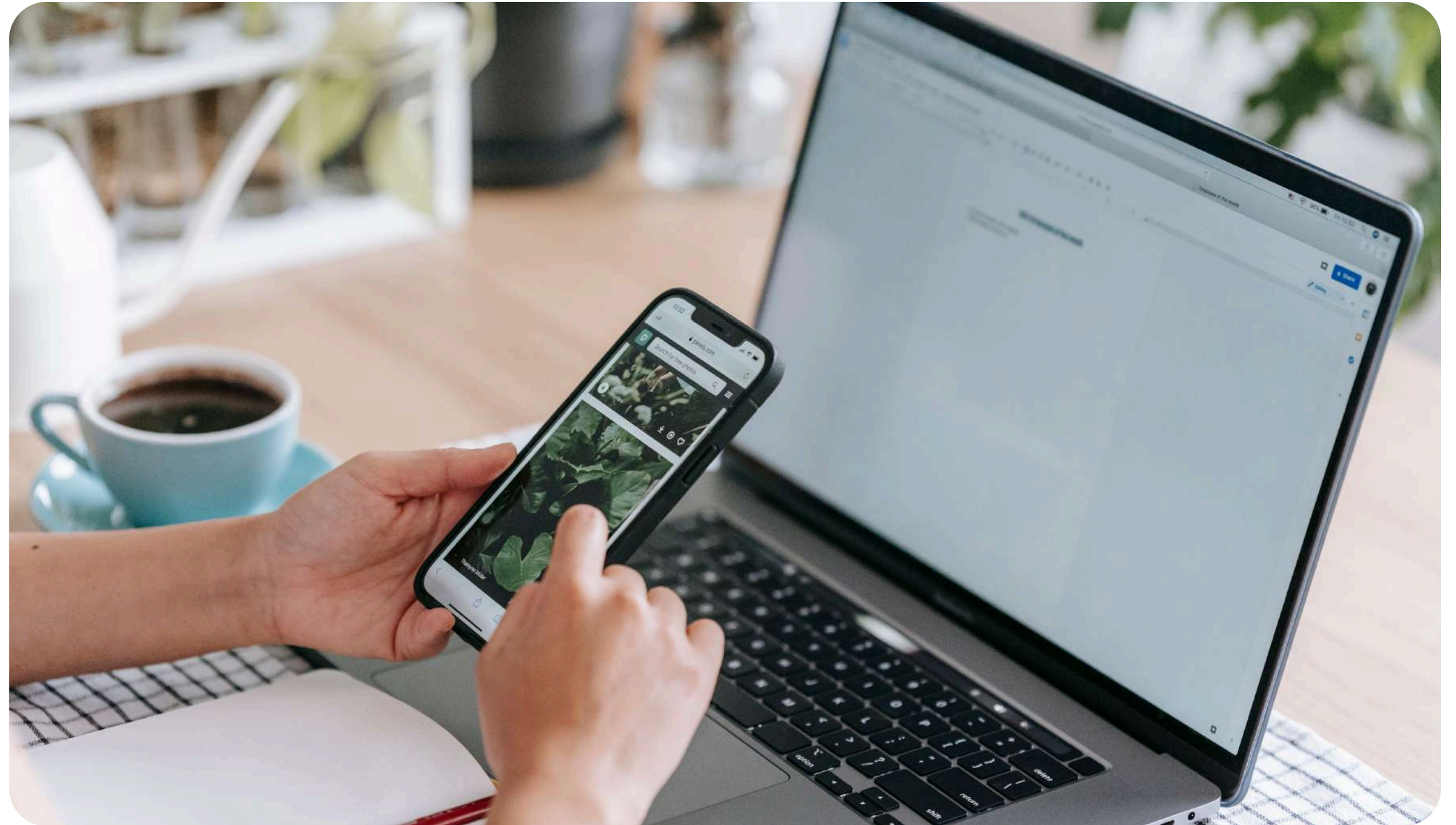
Engineering uses that
knowledge to solve real-
world problems.

Focuses on designing,
building, and improving
products or systems.

Asks "How can we use
this?"

Creates useful machines,
devices, software, and
structures.

- # Strong programming knowledge (C, C++, Python, Java, etc.)
- # Problem-solving and logical thinking
- # Understanding of data structures and algorithms
- # Version control tools (Git and GitHub)
- # A laptop/computer with a good internet connection
- # Communication and teamwork skills
- # Continuous learning to keep up with new technologies

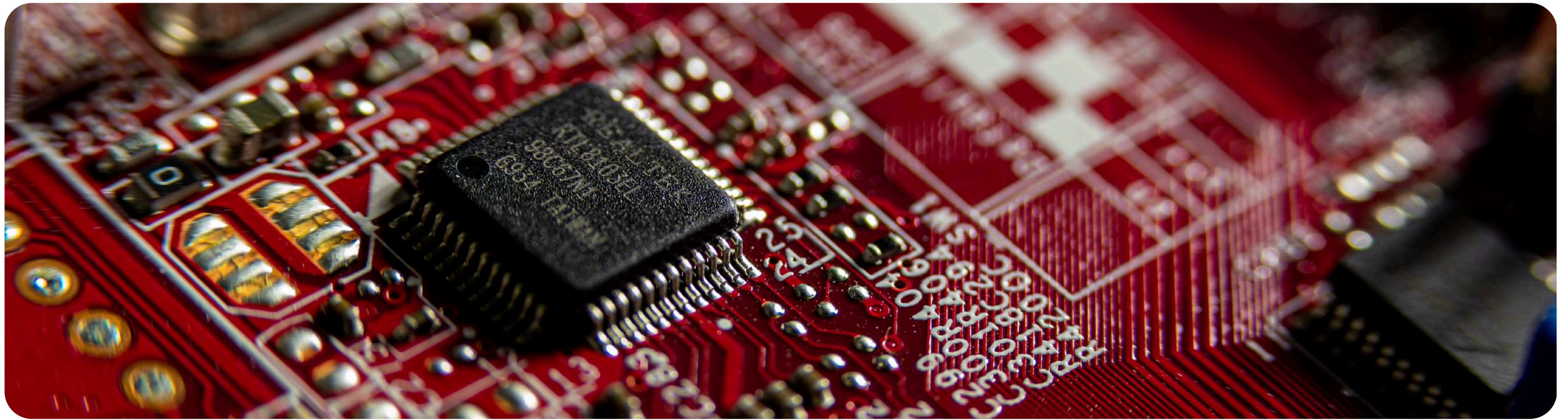


NEEDS OF CSE

WANTS OF CSE

- These help a Computer Science Engineer grow faster and stand out:
- High-paying job in a top tech company
- Knowledge of Artificial Intelligence (AI) and Machine Learning (ML)
- Cybersecurity and ethical hacking knowledge
- Mobile or web development skills
- Professional certifications
- Internships and industry experience
- Work-life balance and career growth opportunities





ROLE OF CSE IN OTHER FIELDS

Civil Engineering

Structural analysis software, Building Information Modeling (BIM), traffic management

Smart City, Smart Traffic Control System

Mechanical Engineering

robotics, automation, simulation

Self-driving cars, Industrial robots

Electrical Engineering

Embedded systems, IoT, smart grid monitoring

Smart Power Grid, Smart Energy Meter

Biomedical Engineering

Medical image processing, AI diagnosis, health monitoring

Wearable Health Monitor, AI-based Disease Detection

CASE STUDY



**Engineering – Solar-
Powered Water Pump for
Farmer**

ENGINEERING TO SOLVE WATER SUPPLY PROBLEMS IN AGRICULTURE

PROBLEM: FARMERS IN A VILLAGE DEPENDED ON DIESEL-POWERED WATER PUMPS FOR IRRIGATION. DIESEL WAS EXPENSIVE, CAUSED POLLUTION, AND WAS NOT ALWAYS AVAILABLE, MAKING IRRIGATION DIFFICULT.

ENGINEERING SOLUTION: A TEAM OF ENGINEERS DESIGNED AND INSTALLED A SOLAR-POWERED WATER PUMPING SYSTEM. THE SYSTEM INCLUDED:

SOLAR PANELS TO GENERATE ELECTRICITY.

AN ELECTRIC WATER PUMP.

A WATER STORAGE TANK.

A CONTROLLER TO MANAGE THE SYSTEM EFFICIENTLY.

IMPLEMENTATION:

SOLAR PANELS WERE INSTALLED NEAR THE FARMLAND.

THE PUMP DREW WATER FROM A BOREWELL.

WATER WAS STORED IN A TANK AND DISTRIBUTED TO CROPS THROUGH PIPES.

